

Vehicle height compounds dangers of speed for pedestrians

New pedestrian injury risk curves calculated by IIHS show that vehicles with taller front ends begin to cause serious injuries at lower speeds.

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The faster a vehicle is moving when it strikes a pedestrian, the more likely it is to inflict serious injuries. Exactly how much more likely depends in part on the height of the vehicle, with taller vehicles compounding the risk from higher crash speeds, [new research](#) from the Insurance Institute for Highway Safety shows.

IIHS researchers analyzed pedestrian crashes to develop injury risk curves showing how speed affects crash outcomes. They found that the effect of crash speed on injury risk was magnified for vehicles with taller front ends. Compared with risk curves developed using crash data from Europe, where tall passenger vehicles are less common, risk curves for the U.S. show pedestrians here begin to suffer more serious injuries at lower speeds.

“A small increase in crash speed can really ramp up the danger to a pedestrian,” IIHS President David Harkey said. “Our fondness for tall SUVs and pickups in the U.S. has intensified that effect.”

The findings suggest that the 25 mph speed limit commonly used in residential neighborhoods in the U.S. may be too high for bustling city centers or other areas with large numbers of pedestrians.

To estimate injury risk at different impact speeds, IIHS researchers examined 202 crashes involving pedestrians ages 16 or older. The records came from two databases — one of crashes that took place between 2015 and 2022 in Michigan and another of crashes that occurred in 2022 in California, New Jersey and Texas.

The researchers used photographs of the vehicle models involved to determine the height of the hood’s front edge. They then looked at the effect of vehicle speed and vehicle height on the risk of injuries of different severities.

They focused on front-end height because that determines the point of impact on a pedestrian’s body, which is a key factor in injury severity. (In contrast, vehicle weight doesn’t play a major role in pedestrian crashes since any vehicle, even a small car, vastly outweighs a person.)

Regardless of vehicle height, higher speeds were associated with increased risk of injury at all severities. Pedestrians struck at 20 mph had a 46% chance of sustaining at least a moderate injury, such as a broken bone or concussion, and an 18% chance of a serious injury, such as a broken bone that protrudes from the skin. At 35 mph, the risk of moderate injury climbed to 86% and the risk of a serious injury rose to 67%. As far as fatalities were concerned, pedestrians struck at 20 mph had only a 1% chance of dying from their injuries, but at 35 mph, the risk reached 19%; at 50 mph, it exceeded 80%.

Pedestrian injury risk by vehicle speed

Roll over the graph to see the risk of injuries of different severities at any given speed.

In general, higher vehicle front ends increased the likelihood of both moderate and serious pedestrian injuries. At 27 mph, the average speed of all 202 crashes, a median-height car had a 60% chance of causing moderate injuries to a pedestrian and a 30% chance of causing serious injuries. In comparison, a median-height pickup — with a front end 13 inches higher than that of a median car — had an 83% chance of causing moderate injuries and a 62% chance of causing serious injuries.

This tracks with earlier IIHS research that found that vehicles with taller front ends are more likely to kill people when they hit them. In the new study, fatalities followed the same pattern as moderate and serious injuries, but the effect wasn’t statistically significant.

The study’s authors compared the risk curves they generated for the effect of speed with risk curves published in 2022 based on crashes in Germany. They found that pedestrians in the U.S. are more likely to be injured at all speeds. Hypothesizing that differences in vehicle size were to blame, they used the U.S. data to generate new injury curves for the median-height pickup and median-height car. The car curves are close to the German curves, while the pickup curves are shifted left, indicating increased risk at lower speeds.

Pedestrian injury risk for median U.S. pickup and median U.S. car vs. German vehicles

Roll over the graphs to see the risk from crashes involving cars and pickups at any given speed.

Moderate injury or worse

Serious injury or worse

Fatality

☐ **Median pickup** ☐ **Median car** ☐ ☐ **German vehicles (Lubbe et al., 2022)**

“Although SUVs are a growing share of the market in Europe, the passenger vehicle fleet there has long been dominated by cars. In contrast, the majority of passenger vehicles on U.S. roads today are SUVs or pickups,” said IIHS Senior Statistician Sam Monfort, the study’s lead author. “These choices have very real consequences for pedestrian safety.”

For serious injuries, the risk curve for the median pickup is not just shifted left compared with the median car, but is steeper as well. In other words, speed increases have a more pronounced effect when taller vehicles are involved. For example, as crash speed increases from 15 mph to 35 mph, the risk of a serious injury goes from 9% to 52% when a median-height car is involved. With a median pickup, the risk shoots up from 11% to 91%.



Passenger vehicles in Germany are generally smaller than in the U.S. That difference shows up in risk curves.

The findings point to the need for policymakers and traffic engineers to account for the makeup of the U.S. fleet when thinking about speed limits and traffic calming.

A commonly used benchmark for a “safe” crash speed is one at which the risk of serious injury is 10% or less. In this study, that threshold occurred at about 15 mph. That means even a 25 mph limit may be too high for areas with a lot of pedestrian traffic.

That said, small reductions above that level could have enormous benefits for pedestrians too. Cutting crash speeds from 30 mph to 25 mph, for example, would cut serious injury risk by more than a third, from 50% to 32%.

Vehicle manufacturers also have a role to play in preventing pedestrian crashes and mitigating those that still occur. They could, for example, modify their front-end designs and equip vehicles with more robust automatic emergency braking systems.

“This study is a vivid illustration of how multiple factors — in this case speed and vehicle height — converge to create negative outcomes on the road,” Harkey said. “Similarly, it will take a combination of actions from different corners of the transportation world to improve pedestrian safety.”

Additional information

A modern injury risk curve for pedestrian injury in the United States: the combined effects of impact speed and vehicle front-end height

Research paper

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